ZOO202 MID TERM OBJECTIVE BEST OVER ALL FILES BY

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BS-ZOOLOGY 4TH SEMESTER

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1. Excretory structures in the kidneys are called **nephrons**.
2. Marine fishes environment contains **3.5% ions** and tissues contain **0.65%**
3. Up to 90% of nitrogenous wastes are eliminated as **ammonia.**
4. In Amphibians lineage is called the **nonamniote lineage.**
5. In Amphibians Skull i**s dicondylic.**
6. **Order Labyrinthodontia** Also known as "Stem Amphibia"
7. **Order Phyllospondyli** Small Salamander like
8. **Order Lepospondyli** Small eel like.
9. In salamanders, **30 to 90%** of gas exchange occurs across the skin.
10. **At 5° C,** approximately 70% of gas exchange occurs across the skin and mouth lining of a frog.
11. Critical temperature extremes for some salamanders lie between **2 and 27°C**, and for some anuransbetween **3 and 41° C.**
12. Pectoral girdle to the inner ear through a second ossicle, called the **operculum.**
13. Anurans use **low- and high-frequency sounds** in different situations.
14. Fishes are members of the chordate subphylum **Vertebrata.**
15. Shark teeth are actually modified **placoid** scales.
16. The largest sharks are not predatory but are **filter feeders.**
17. Herring, paddlefishes, and whale sharks are **filter feeders.**
18. sturgeons are valued for their **caviar (eggs**),
19. Lamprey, are **external parasites.**
20. Lampreys are **agnathans** in the class **Cephalaspidomorphi**
21. **Paddlefishes** are large, freshwater chondrosteans.
22. Animals in the phyla Hemichordata and Chordata share deuterostome characteristi with
23. **echinoderms.**
24. Sharks, rays, skates, and ratfishes are members of **Class Chondrichthyes**
25. **Hagfishes** are the most primitive vertebrates known.
26. Lungfishes and Coelacanth belong to **Class Sarcopterygii**
27. The organisms belonging to the Protochordata are generally known as the **lower chordates**.
28. **Deuterostomate** animals with a body divided into three regions: proboscis, collar, and trunk;
29. **Enteropneusta** Food is digested as diverticula of the gut, called hepatic sacs, relea enzymes.
30. The first mammalian fossils date back to the Triassic period, about **208** million ye ago.
31. Finally,bird fossils appear **146** million years ago in the Jurassic period.
32. A Precambrian fossil known **as *Yarnemia*has** been referred to the Urochordata,
33. The body wall of most tunicates (L. tunicatus, to wear a tunic or gown) is a connective-tissue-likecovering, called the **tunic**,

34. Instead of blood vessels, the heart pumps blood through a number of small pockets called **sinuses.**

35. The majority of the organs lie with in a body cavity below the atrial cavity called t **epicardium.**

36. Tunicate **blood plasma is colorless** and contains various kinds of amoeboid cells.

37. Members of the subphylum Cephalochordata are called **lancelets**.

38. The cephalochordates consist of two genera, **Branchiostoma(amphioxus)** and **Asymmetron**, andabout 45 species.

39. Larger materials catch on cilia of the **cirri .**

40. A diverticulum off the gut, called the **midgut cecum**, extends anteriorly .

41. Recent cladistic analysis of vertebrate evolution indicates that a group of fishes called **hagfishes arethe most primitive vertebrates** known (living or extinct).

42. A group of ancient eel-like animals, **the conodonts**, are known from fossils that date back about **510million years.**

43. It was present in the bony armor of a group of fishes called **ostracoderms**.

44. The importance of freshwater in the evolution of fishes is evidenced by the fact th over **41% of allfish species are found in freshwater**.

45. Even though freshwater habitats represent only a small percentage **(0.0093% by volume)** of theearth's water resources.

46. **Hagfishes** are members of the class Myxini

47. **Lampreys**are agnathansin the class Cephalaspidomorphi.

48. Ostracodermswere **bottom dwellers**, often about **15 cm long**.

49. Lampreys have **salivary glands** with anticoagulant secretions and feed mainly on the blood of theirprey.

50. Members of the genus Lampetraare called **brook lampreys**.

51. Ammocoetelarvae grow from **7 mm to about 17 cm** over three years.

52. Adults return only once to the headwaters of their stream to **spawn and die**.

53. Sharks arose from early jawed fishes midway through the Devonian period, about **375 million yearsago**.

54. A second major group of chondrichthians, in the subclass **Holocephali**contains ab 30 species .

55. The first fossils of bony fishes are from **late Silurian** deposits (approximately 405 million years old).

56. One group of sarcopterygians are the **lungfishes**.

57. **Lungfishes** may remain in aestivation for six months or more.

58. second group of sarcopterygians are the **coelacanths**.

59. A third group of sarcopterygians, called **osteolepiforms**, became extinct before the close of thePaleozoic period.

60. The subclass Actinopterygii contains fishes that are sometimes called the **ray finned fishes** becausetheir fins lack muscular lobes.

61. The sturgeon mouth is small, and **jaws are weak**.

62. Most **sturgeons**live in the sea and migrate into rivers to breed.

63. **Paddlefishes**are large, freshwater chondrosteans.

64. **Amiais**commonly referred to as the **dogfish** or bowfin.

65. The blood secretes gases into the swim bladder in a vascular network called the **re mirabile**.

66. IN Protochordata, At a certain stage of their lives, their body develops a long, rod-like structure for support called **notochord.**

67. In Hemichordata, Primitive notochord restricted to proboscis only, thus called **stomochord.**

68. In Acorn worms, Pharyngeal slits are openings between the anterior region of the digestive tract, **called the pharynx.**

69. In Enteropneusta: Food is digested as diverticula of the gut, called **hepatic sacs.**

70. At low tide, coils of fecal material, **called castings**, lie on the substrate at burrow openings.

71. In Enteropneusta, nervous system Lies at the base of the ciliated epidermis. Consi: of dorsal and ventral nerve tracts and a network of epidermal nerve cells, **called a nerve plexus**.

72. All blood flowing anteriorly passes into a series of blood sinuses, called **the glomerulus**, at the base of the proboscis.

73. The body wall of most tunicates (L. tunicatus, to wear a tunic or gown) is a connective-tissue-like covering**, called the tunic**, that appears gel-like but is often quite tough.

74. Rootlike extensions of the tunic, called **stolons**, help anchor a tunicate to the substrate and may connect individuals of a colony.

75. Tunicates derive their name from a thick extracellular body covering, called **a tun** that is made up of cellulose-like, complex sugars.

76. the heart pumps blood through a number of small pockets **called sinuses.**

77. The adult tunicate body is shaped like a small barrel with two large openings calle **siphons.**

78. The majority of the organs lie within a body cavity below the atrial cavity called tl **epicardium.**

79. The most obvious internal structures of the urochordates are a very large pharynx and a cavity, called **the atrium,** that surrounds the pharynx laterally and dorsally.

80. Numerous pharyngeal slits called **stigmas** perforate the pharynx.

81. During feeding, cells of a ventral, ciliated groove, called the **endostyle**, form a mucous sheath.

82. Members of the subphylum Cephalochordata are called **lancelets.**

83. Ciliated, fingerlike projections, called **cirri,** hang from the ventral aspect of the or hood and are used in feeding.

84. The opening from the atrium to the outside is called the **atriopore**

85. A diverticulum off the gut, called the **midgut cecum**, extends anteriorly.

86. Recent cladistic analysis of vertebrate evolution indicates that a group of fishes called **hagfishes** are the most primitive vertebrates **known (living or extinct).**

87. It was present in the bony armor of a group of fishes called **ostracoderms.**

88. Members of the genus Lampetra are called **brook lampreys.**

89. A narrow, tapering tail has resulted in the common name **"ratfish."**

90. Since that time, specializations not found in other elasmobranchs have evolved, including a gill cover, called **an operculum.**

91. A third group of sarcopterygians, **called osteolepiforms**, became extinct before th close of the Paleozoic period.
92. The subclass Actinopterygii contains fishes that are sometimes **called the ray fin fishes** because their fins lack muscular lobes.
93. A vessel to the lungs has developed as a branch **off aortic arch VI** This vessel is called the **pulmonary artery.**
94. The atrium and ventricle are **partially divided.**
95. Maintain water flow by holding their mouths open. This method is called **ram ventilation.**
96. Gill filaments extend from each gill arch **called pharyngeal lamellae.**
97. Teleosts have **swim bladders.**
98. The blood secretes gases into the swim bladder in a vascular network called **the re mirabile ("miraculous net").**
99. Excretory structures in the kidneys are **called nephrons.**
100. Male elasmobranchs have modified pelvic fins called **claspers.**
101. A second lineage flourished into the Jurassic period. This lineage is called t **nonamniote lineage.**
102. Members of the family Salamandridae are commonly **called newts.**
103. Supportive processes called **zygapophyses** on each vertebra prevent twistin
104. The first vertebra is a **cervical vertebra**
105. The last trunk vertebra is a **sacral vertebra**
106. Sternum is present in the **anterior ventral trunk region**
Unlike other vertebrates, amphibians have contractile vessels**, called lymphatic hearts.**
107. Gas exchange across the skin is called **cutaneous respiration.**
108. The forebrain contains **olfactory centers and regions** that regulate color change and visceral functions.
109. The midbrain contains a region called **the optic tectum.**
110. The midbrain also processes **visual sensory information.**
111. The lower eyelid is movable, and cleans and protects the eye. Much of it is transparent and is called the **nictitating membrane.**
112. a fold of epithelium called the **iris surrounds it.**
113. Abutting the tympanic membrane is a middle-ear ossicle (bone) called the **stapes (columella)**
114. Pectoral girdle to the inner ear through a second ossicle, called the **operculum.**
115. Amphibian development includes larval stages called **tadpoles**. Tadpoles often differ from the adults.
116. The male grasps the female in a position **called amplexus.**
117. The skin of amphisbaenians has ringlike folds called **annuli.**

118. A plate of bone, called the **secondary palate**, evolved in the archosaurs and separates the nasal and mouth passage ways.

119. All reptiles periodically shed the outer, epidermal layers of the skin in a process called **ecdysis.**

120. The organisms belonging to the Protochordata are generally known as **the lower chordates.**

121. Protochordata are also known **as Acraniata** because they lack a true skull.

122. A Precambrian fossil known as **Yarnemia** has been referred to the Urochordata, but this assignment is doubtful.

123. Urochordates are also known **as tunicates** because their body is surrounded a leathery sheath composed of tunicin (cellulose).

124. A group of ancient eel-like animals, the **conodonts,** are known from fossils that date back about **510 million years.**

125. Muscle blocks along the body wall suggest an active swimming existence. This evidence means that **these 530 million-year-old animals located prey by si** and then pursued it through prehistoric seas.

126. Oldest vertebrate fossils: Jawless fish were only vertebrates for more than **5 m.y.a**

127. They are found mainly in lakes and large rivers of the **Mississippi River** ba and are also known **from western North America and China**

128. Order Labyrinthodontia Also known **as "Stem Amphibia.**

129. Root like extensions of the tunic, called **stolons**,